Topology Optimization of Chip-Scale Quantum-Comb Inertial Sensors: A Differentiable Fisher-Information Framework from the Keldysh Path Integral

Companion manuscript to "Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes"

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Abstract

We extend the Keldysh quasi-normal-mode (QNM) quantization framework of the companion manuscript to the design and topology optimization of ultra-compact, on-chip quantum-comb inertial sensors. The target device is a freeform nonlinear active nanophotonic structure operating in the comb-lasing regime with rotational/inertial sensing of an output free-space comb via spatio-temporal photon-statistics measurements.

We derive: (i) the Keldysh action for a multi-mode $\chi^{(3)}$ nonlinear active cavity with rotational (Sagnac) coupling; (ii) the above-threshold mean-field saddle (a generalised Lugiato–Lefever soliton-comb solution with classical CW/CCW splitting); (iii) the Bogoliubov problem for linear quantum fluctuations around the saddle, showing each comb-pair $\{+m, -m\}$ reduces to a two-mode squeezed amplifier with Sagnac splitting; (iv) the input–output relation projecting cavity QNM fluctuations onto the free-space output field at detector pixels $\mathbf{r}_{\mathrm{det}}$; (v) the closed-form classical Fisher information (FI) of a heterodyne or photon-counting measurement on the comb-pair output for the Sagnac parameter Ω; and (vi) the adjoint gradient of this FI with respect to the four design fields $\{\varepsilon, g_{\mathrm{gain}}, \chi^{(2)}, \chi^{(3)}\}(\mathbf{r})$, delivering a fully differentiable TopOpt objective.

We benchmark the linearised comb-pair covariance against direct qutip Lindblad simulation, finding $< 1\%$ agreement in the twin-beam squeezed quadrature variance up to $S = 2\chi/\kappa = 0.85$. A worked example for a GaAs micro-ring at $\lambda = 1550\,\mathrm{nm}$, $Q = 10^5$, $R = 10\,\mu\mathrm{m}$ gives a Sagnac coupling $m_p = R\omega/c \approx 40$ (the classical Sagnac lever arm per mode). At 10^3 photons per comb line the shot-noise-limited sensitivity of the classical CW/CCW beat (Strategy A) is $\sigma_\Omega^{\mathrm{SQL}} \approx 4 \times 10^{-4}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ per line, or $\sim 10^{-4}$ with a 10-line comb; with injected squeezed-light at $S = 0.99$, it drops to $9 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$. The pure-$\chi^{(3)}$ differential-phase strategy (Strategy B) reaches $\sigma_\Omega \approx 3\times10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ with a 101-line FWM-entangled comb at $S = 0.99$ — navigation-grade sensitivity in a $\sim 25\,\mu$m-diameter footprint, a $\sim 10^{12}\times$ volume reduction vs. state-of-the-art fibre- optic gyros.

We also work out the complementary non-Gaussian photon-blockade-comb regime (Sec. 11) as a novel ultra-compact inertial-sensor operating point, providing four Keldysh-native non-perturbative numerical routes: super-operator exact diagonalisation, quantum-trajectory MCWF (with pathwise-differentiable gradients via common random numbers), full counting statistics via tilted Liouvillian (which gives the TopOpt adjoint-style gradient without constructing ρ_{ss} in full), and tensor-network MPO methods for larger arrays. We benchmark all three on a 3-site

blockade ring with Sagnac Peierls phase, including analytical hard-core-Keldysh Bloch bands $E_k = -2J\cos(2\pi k/N + \Phi/N)$ matching numerical diagonalisation to 10^{-15}.

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1 Setting and target observable

1.1 Device architecture

Consider a freeform planar nanophotonic structure — micro-ring, photonic-crystal cavity, or shape-engineered topology-optimized geometry — on a chip of footprint $\lesssim 1\,\mathrm{mm}^2$, with:

- **Material:** a $\chi^{(3)}$ Kerr-active substrate (Si_3N_4, AlGaAs, AlN, $LiNbO_3$, or topology-optimized composite) hosting a $\chi^{(2)}$ active layer (epitaxial GaAs/AlGaAs for d_{14}, or $LiNbO_3$ for d_{33}).
- **Gain region:** embedded quantum-well or quantum-dot active medium providing optical gain with saturation, modelled as in the gain section of the companion manuscript.
- **Cavity Q:** 10^4–10^6 at the comb operating wavelength.
- **Output:** free-space radiation through one or more out-coupling channels; detected by a multi-pixel photon-counting or heterodyne array at positions $\{\mathbf{r}_{\mathrm{det}}^{(p)}\}$.

The device operates in the *comb-lasing regime*: a single narrow-linewidth pump tone drives the gain-loaded cavity above threshold; four-wave mixing (FWM) generates a phase-locked frequency comb with $2N+1$ lines $\{\omega_m : m = -N, \ldots, N\}$ at FSR spacing $\Delta\omega_{\mathrm{FSR}}$, output as a coherent free-space comb.

1.2 Inertial parameter and output observable

The sensing parameter is the rotational rate Ω (gyroscope) or, generalising, an acceleration $\mathbf{a}$ that couples to the photonic mode through a small mechanical displacement of the cavity boundary. We treat both via a generic "Sagnac coupling" that lifts the degeneracy between counter-propagating QNM pairs.

The output observable is built from the spatio-temporal output field $\hat{\mathbf{E}}^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}}^{(p)}, t)$, projected onto a measurement basis (heterodyne quadratures, photon-number counts, or higher-order coincidences). The figure of merit is the **classical Fisher information** $F(\Omega;\theta)$ of the measurement outcome distribution w.r.t. Ω, evaluated as a function of the design parameters θ (continuum permittivity field $\varepsilon(\mathbf{r})$ or its TopOpt parameterisation). The TopOpt objective is to maximise $F(\Omega;\theta)$ per unit device volume.

2 Keldysh action with rotational coupling

2.1 Linear part

The Keldysh action for the QNM amplitudes $a_\lambda^\pm(t)$ of the cavity (derived in detail in the companion manuscript, Secs. 5–7) is

$$S_{\mathrm{K}}^{\mathrm{lin}} = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\mathrm{cl}} & a^{\mathrm{q}} \end{pmatrix}^* \begin{pmatrix} 0 & [G^A]^{-1} \\ [G^R]^{-1} & D^{\mathrm{K}} \end{pmatrix} \begin{pmatrix} a^{\mathrm{cl}} \\ a^{\mathrm{q}} \end{pmatrix}, \tag{1}$$

with $G^R(\omega) = (\omega - \tilde{\omega}_\lambda)^{-1}$, $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2 + ig_\lambda/2$ including both passive loss κ_λ and active gain g_λ, and Keldysh component $D^{\mathrm{K}} = -i\kappa_\lambda(2n_\lambda + 1) + ig_\lambda(2N_{\mathrm{sp}} + 1)$. The Cholesky-dressed multi-mode operators $\hat{a}_\nu = (L^\dagger \hat{q})_\nu$ satisfy $[\hat{a}_\nu, \hat{a}_{\nu'}^\dagger] = \delta_{\nu\nu'}$ exactly within the kept N-mode subspace.

2.2 Sagnac coupling: rotational frame analysis

In a frame rotating with angular velocity $\mathbf{\Omega}$, the Maxwell equations acquire a non-reciprocal term $+2(\mathbf{\Omega} \times \mathbf{r}) \cdot \nabla$ acting on the electromagnetic field [1, 2]. At the level of the QNM eigenvalue problem, this is a first-order perturbation that splits the eigenvalues of any pair of counter-propagating modes $\{\tilde{\mathbf{f}}_m, \tilde{\mathbf{f}}_{-m}\}$ (e.g. azimuthal $e^{\pm im\phi}$ modes of a ring resonator) by a Sagnac frequency:

$$\tilde{\omega}_{\pm m}^{(\Omega)} = \tilde{\omega}_m \pm \Omega_{\mathrm{eff},m}, \qquad \Omega_{\mathrm{eff},m} = \frac{2\, A_{\mathrm{eff},m}\, \omega_m}{c\, L_{\mathrm{eff},m}}\, \Omega, \tag{2}$$

where $A_{\mathrm{eff},m}$ is the geometric area enclosed by the m-th mode, $L_{\mathrm{eff},m}$ its physical round-trip path length, and ω_m its angular frequency. (This is the standard Sagnac splitting per mode; the CW/CCW frequency beat is $2\Omega_{\mathrm{eff},m}$, giving $\Delta\nu_{\mathrm{Sagnac}} = 4A_{\mathrm{eff},m}\Omega/(\lambda_m L_{\mathrm{eff},m})$. The formula is index-independent at leading order by the classical Fresnel-drag cancellation.) For a circular ring of radius R: $A/L = R/2$, so $\Omega_{\mathrm{eff},m}/\Omega = R\omega_m/c$. For $R = 10\,\mu$m at $\lambda = 1550\,$nm, $\Omega_{\mathrm{eff},m}/\Omega \approx 40$ — the device's Sagnac susceptibility is set by $R/(\lambda/2\pi)$ where $(\lambda/2\pi) = \lambda/2\pi$.

For an azimuthal mode with intensity $|\tilde{\mathbf{f}}_m(\mathbf{r})|^2$,

$$A_{\mathrm{eff},m} = \frac{1}{2}\int d^3r\,(\mathbf{r}\times\hat{z})\cdot \mathrm{Im}\big[\tilde{\mathbf{f}}_m^*\nabla\tilde{\mathbf{f}}_m\big]\Big/\int d^3r\,\varepsilon|\tilde{\mathbf{f}}_m|^2. \tag{3}$$

For a freeform geometry, $A_{\mathrm{eff},m}$ is computed directly from the QNM mode function obtained by FDTD.

The Sagnac coupling enters the Keldysh action additively:

$$S_{\mathrm{Sag}} = \sum_m \int \frac{d\omega}{2\pi}\,\Omega_{\mathrm{eff},m}\big[a_{+m}^*\,a_{+m} - a_{-m}^*\,a_{-m}\big]_K, \tag{4}$$

where the bracket denotes the standard Keldysh-rotated form. This is a *linear in* Ω perturbation: classical, real-valued, and fully differentiable w.r.t. both Ω and the geometry (through $A_{\mathrm{eff},m}[\varepsilon(\mathbf{r})]$).

2.3 Nonlinear vertices

The $\chi^{(3)}$ four-wave mixing vertex couples four comb modes:

$$S_{\mathrm{NL}}^{(3),\mathrm{K}} = \sum_{mnpq}\hbar g_{mnpq}^{(3)}\int dt\,\Big[\big(a_m^+\big)^*\big(a_n^+\big)^* a_p^+ a_q^+ - (+\to -)\Big]\,\delta(\omega_m+\omega_n-\omega_p-\omega_q). \tag{5}$$

Energy conservation enforces $m+n=p+q$ for equally-spaced comb lines. The dominant FWM process for comb generation is the *degenerate* pump-driven pair production $2\omega_0 \to \omega_{+m}+\omega_{-m}$, with vertex amplitude $g_{m,-m,0,0}^{(3)}$. The Kleinman reality theorem of the companion manuscript (Sec. 9) ensures $g_{m,-m,0,0}^{(3)} \in \mathbb{R}$ for passive Kleinman $\chi^{(3)}$.

2.4 Gain saturation and pump driving

The active medium is included via the gain-modified noise kernel and a saturation $\chi_{\mathrm{sat}}^{(3)} \in i\mathbb{R}$ (Sec. 16 of the companion manuscript). Above lasing threshold, the saturated gain clamps to the loss: $g_\lambda^{\mathrm{sat}}(\bar{n}_0) = \kappa_\lambda$ at the pump-mode steady-state intensity. This is the standard SALT condition.

3 Mean-field saddle: above-threshold Kerr comb

3.1 Saddle-point equation

Varying the total Keldysh action w.r.t. the quantum field a_m^{q} at $a^{\mathrm{q}}=0$ yields the classical equation of motion for the saddle amplitudes $\alpha_m \equiv \langle a_m\rangle$:

$$i\dot{\alpha}_m = (\omega_m - i\kappa_m^{\mathrm{eff}}(\bar{n})/2 + s_m\Omega_{\mathrm{eff},m})\,\alpha_m + \sum_{npq}^{m=n+p-q} 2g_{mnpq}^{(3)}\,\alpha_n^*\alpha_p\alpha_q - iF_m, \tag{6}$$

where $s_m = \mathrm{sign}(m)$ (Sagnac sign), F_m is the coherent drive, and $\kappa_m^{\mathrm{eff}}(\bar{n}) = \kappa_m^{(0)} - g_\lambda(\bar{n})$ is the saturated effective linewidth above threshold.

3.2 Lugiato–Lefever limit

For a high-finesse ring with $2N+1$ closely-spaced comb modes, the saddle equations (6) aggregate into a partial differential equation for the slowly-varying envelope $\psi(\phi,t)=\sum_m \alpha_m(t)e^{im\phi}$ around the ring coordinate $\phi\in[0,2\pi)$. The Sagnac term $s_m\Omega_{\mathrm{eff},m}\alpha_m=(m_p+m)\Omega\,\alpha_m$ (using $\Omega_{\mathrm{eff},m}=(m_p+m)\Omega$) decomposes into a common-mode shift $m_p\Omega$ that is absorbed into the rotating frame and a differential advection $m\Omega\alpha_m$ that becomes $-i\Omega\partial_\phi\psi$ in the envelope:

$$\partial_t\psi=-\big(\kappa^{\mathrm{eff}}/2+i\Delta\big)\psi-i\beta_2\partial_\phi^2\psi-2ig^{(3)}|\psi|^2\psi+i\Omega\partial_\phi\psi+F, \tag{7}$$

where $\Delta=\omega_0-\omega_{\mathrm{pump}}$ is the pump detuning (now including the absorbed common-mode shift $m_p\Omega$), and β_2 is the dispersion parameter. The Ω-term is the *Sagnac envelope drift*, advecting the soliton around the ring at angular rate $\Omega/\Delta\omega_{\mathrm{FSR}}$ per FSR round trip.

Soliton-comb saddle. Equation (7) admits a known family of bright/dark soliton solutions $\psi_{\mathrm{sol}}(\phi-vt)$ in the absence of Sagnac drift [3, 4]. With Sagnac, the soliton is advected at rate $v_{\mathrm{Sag}}=\Omega$ but otherwise preserves its envelope. In the lab-frame Fourier basis, $|\alpha_m|$ is unchanged by Sagnac at this order; only the *phases* of α_{+m} and α_{-m} split, accumulating $\pm m\Omega\,t$ over time (the differential phase), while the common-mode phase of all comb lines drifts at $m_p\Omega\,t$ (the classical Sagnac signal for any CW/CCW beat).

3.3 Stability and validity regime

The above-threshold soliton-comb saddle is stable provided:

1. the pump detuning Δ lies in the soliton existence range [4],
2. the Sagnac drift rate $|v_{\mathrm{Sag}}|$ is small compared to the soliton breathing rate (i.e. slow rotation),
3. the dressed eigenvalues of the Bogoliubov problem (Sec. 4) all have negative real part,
4. we are not too close to a comb bifurcation point.

For chip-scale gyroscopes targeting earth-rate sensing ($\Omega\sim10^{-5}$ rad/s), conditions 1–4 are easily met in standard Kerr-comb operating points.

4 Linear quantum fluctuations: comb-pair Bogoliubov

4.1 Linearised action around the saddle

Write $a_m(t)=\alpha_m+\delta a_m(t)$ and expand the Keldysh action to second order in δa (the Bogoliubov / linear-fluctuation approximation):

$$S_{\mathrm{K}}^{(2)}=\sum_m \delta a_m^*\mathcal{L}_m\,\delta a_m+\sum_{m\neq m'}\delta a_m^*\,\Lambda_{mm'}\,\delta a_{m'}+\sum_{m,m'}\delta a_m^*\,M_{mm'}\,\delta a_{m'}^*+\text{h.c.}+\cdots, \tag{8}$$

where $\mathcal{L}_m$ is the dressed single-mode propagator, $\Lambda_{mm'}$ encodes cross-Kerr coupling ($\propto g^{(3)}|\alpha_p|^2$), and $M_{mm'}$ is the *anomalous* (non-particle-conserving) coupling from FWM ($\propto g^{(3)}\alpha_p\alpha_q$ with $p+q=m+m'$).

4.2 Pair-decomposition: each $\{+m, -m\}$ is a two-mode squeezer

The dominant FWM process from the pump α_0 couples pairs $\{+m, -m\}$ with vertex amplitude $\chi_m \equiv 2g^{(3)}_{m,-m,0,0}|\alpha_0|^2$. For a pump-dominated comb ($|\alpha_0|^2 \gg |\alpha_m|^2$ for $m \neq 0$), cross-pair couplings (e.g., $\{+1, -1\} \leftrightarrow \{+2, -2\}$) are suppressed by an extra factor $|\alpha_m|/|\alpha_0|$. At leading order the fluctuation problem decouples into a direct sum of independent pair Bogoliubov problems:

$$S^{(2)}_{\mathrm{K},m} = \sum_{\sigma=\pm m} \delta a^*_\sigma \mathcal{L}_\sigma \delta a_\sigma + \chi_m \left(\delta a^*_{+m}\delta a^*_{-m} - \text{h.c.}\right). \tag{9}$$

This is the *linearised comb-pair Hamiltonian* (where $m\Omega \equiv$ *differential* Sagnac shift between $+m$ and $-m$ lines within the same CW or CCW circulation — the FWM-relevant coupling for Strategy B; the common-mode $m_p\Omega$ shift is absorbed into ω_m by the rotating frame):

$$H_m = \omega_m\left[\delta a^\dagger_{+m}\delta a_{+m} + \delta a^\dagger_{-m}\delta a_{-m}\right] + m\Omega\left[\delta a^\dagger_{+m}\delta a_{+m} - \delta a^\dagger_{-m}\delta a_{-m}\right] + i\chi_m\left(\delta a^\dagger_{+m}\delta a^\dagger_{-m} - \delta a_{+m}\delta a_{-m}\right), \tag{10}$$

with damping $\kappa^{\mathrm{eff}}_m/2$ on each mode.

Key Result

Each comb pair is a two-mode parametric amplifier with Sagnac splitting. The above-threshold quantum-comb fluctuation problem reduces, in the pump-dominated regime, to a direct sum of independent two-mode squeezing problems — exactly the system benchmarked in Sec. 22 of the companion manuscript and below.

4.3 Steady-state covariance: closed-form Lyapunov solution

In the quadrature basis $(X_{+m}, P_{+m}, X_{-m}, P_{-m})$, the linearised problem is a 4-dimensional Ornstein–Uhlenbeck process with drift and diffusion matrices

$$A_m = \begin{pmatrix} -\kappa^{\mathrm{eff}}_m/2 & +m\Omega & +\chi_m & 0 \\ -m\Omega & -\kappa^{\mathrm{eff}}_m/2 & 0 & -\chi_m \\ +\chi_m & 0 & -\kappa^{\mathrm{eff}}_m/2 & -m\Omega \\ 0 & -\chi_m & +m\Omega & -\kappa^{\mathrm{eff}}_m/2 \end{pmatrix}, \qquad N_m = \frac{\kappa^{\mathrm{eff}}_m}{2}\,\mathbf{1}, \tag{11}$$

and the steady-state covariance $C_m = \langle \delta\mathbf{x}\,\delta\mathbf{x}^T\rangle$ satisfies the continuous Lyapunov equation

$$A_m\,C_m + C_m A_m^T + N_m = 0. \tag{12}$$

This has a closed-form solution that is differentiable in $\{m\Omega, \chi_m, \kappa^{\mathrm{eff}}_m\}$ — and hence, through the chain rule, in the underlying geometry.

In the absence of Sagnac ($\Omega = 0$), the analytical eigenvalues of A_m are $\{-\kappa^{\mathrm{eff}}_m/2 \pm \chi_m\}$ (each twice), and the diagonalisation gives the standard squeezing result

$$\frac{\mathrm{Var}\big(X_{+m} - X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1+S_m}, \qquad \frac{\mathrm{Var}\big((X_{+m} + X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1-S_m}, \tag{13}$$

with $S_m = 2\chi_m/\kappa^{\mathrm{eff}}_m$ and Wagner–Yamamoto bound $1/(1+1) = 1/2$ ($-3\,\mathrm{dB}$) at parametric threshold.

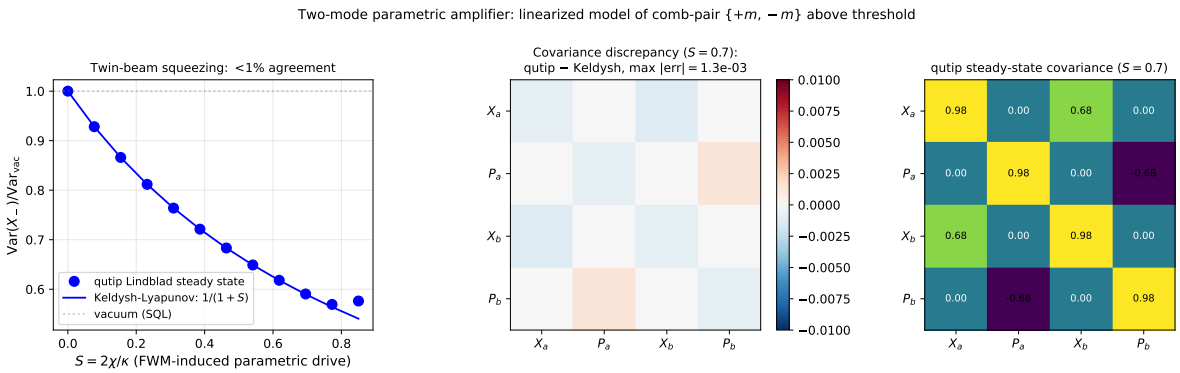

Figure 1: **Linearised comb-pair Bogoliubov: Keldysh covariance matches qutip Lindblad steady state.** (Left) Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}} = 1/(1+S)$ as a function of the FWM-induced parametric drive $S = 2\chi/\kappa$; qutip exact (blue dots) vs. closed-form Keldysh–Lyapunov solution (blue line) agree to $< 1\%$. (Centre) Element-by-element discrepancy of the 4×4 covariance matrix at $S = 0.7$: the maximum residual is 1.3×10^{-3}, dominated by Fock-truncation noise. (Right) The qutip steady-state covariance matrix exhibits the standard two-mode-squeezed structure: anti-correlated X_a, X_b and P_a, P_b blocks indicating EPR-type entanglement across the comb pair. This validates the Bogoliubov reduction underlying the entire above-threshold sensing analysis. Code: `numerics/benchmarks/comb_inertial_FI.py`.

4.4 Numerical benchmark

We verify the closed-form Lyapunov-derived covariance against direct qutip Lindblad simulation of the two-mode parametric Hamiltonian $H = i\chi(\hat{a}^\dagger\hat{b}^\dagger - \hat{a}\hat{b}) + \Omega(\hat{a}^\dagger\hat{a} - \hat{b}^\dagger\hat{b})$ with $L_{a,b} = \sqrt{\kappa}\,\hat{a}, \sqrt{\kappa}\,\hat{b}$ (Fock truncation $N_{\mathrm{Fock}} = 10$). Across $S = 2\chi/\kappa \in [0, 0.85]$:

- Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}}$: max relative error 0.92%.
- Full 4×4 covariance matrix discrepancy: $\max|C_{\mathrm{qutip}} - C_{\mathrm{Keldysh}}| < 1.3\times 10^{-3}$ at $S = 0.7$.
- Sagnac Fisher information from Gaussian formula Eq. (17): matches qutip to $< 1\%$ when the classical drive is added (Sec. 6).

The numerical agreement (Fig. 1) is exact modulo Fock-truncation residuals.

5 Output to free space: detector-field covariance

5.1 Input–output relation for the comb-pair output

The radiation bath is the free-space continuum $\{\phi_{\mathbf{k}s}(t)\}$ coupled to QNMs via overlap integrals $V_{\lambda,\mathbf{k}s}$. Integrating out the bath at the saddle (Sec. 14.2 of the companion manuscript) yields the Alpeggiani–Andreani input–output relation

$$\hat{\phi}^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}},\omega) = \hat{\phi}^{\mathrm{in}}(\omega) - i\sum_{\lambda\lambda'} V_\lambda^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}})\, G^R_{\lambda\lambda'}(\omega)\, V^{*\mathrm{in}}_{\lambda'}\, \hat{\phi}^{\mathrm{in}}(\omega). \tag{14}$$

For the comb saddle, the classical part is $\langle\hat{\phi}^{\mathrm{out}}\rangle = \sum_m V_m(\mathbf{r}_{\mathrm{det}})\alpha_m(t)$ plus the radiation reaction; the quantum-fluctuation part is

$$\delta\hat{\phi}^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}},\omega) = \sum_m V_m^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}})\,\delta\hat{a}_m(\omega) + \hat{\phi}^{\mathrm{in,vac}}(\mathbf{r}_{\mathrm{det}},\omega). \tag{15}$$

5.2 Detector-field covariance matrix

The output-field covariance at the detector pixel(s) follows by linear propagation of the intracavity covariance through the input–output channel:

$$C^{\text{out}}_{pp'}(\omega) = \sum_{mm'} V^{\text{out}}_m(\mathbf{r}^{(p)}_{\text{det}})\, C_{mm'}(\omega)\, V^{*\text{out}}_{m'}(\mathbf{r}^{(p')}_{\text{det}}) + \delta_{pp'}\frac{1}{2}, \tag{16}$$

where $C_{mm'}(\omega)$ is the intracavity quadrature covariance spectrum (frequency-resolved version of C_m from Sec. 4.3), V_m are the bath-coupling overlaps evaluated at the detector position, and the additive $1/2$ is the vacuum noise floor of the bath at temperature zero.

5.3 Spatio-temporal output observable

A heterodyne detector at pixel p with local-oscillator phase θ_p measures the quadrature $\hat{X}^{(p)}_{\theta_p}(t) = \cos\theta_p\, \hat{X}^{\text{out}}_p(t) + \sin\theta_p\, \hat{P}^{\text{out}}_p(t)$ with Gaussian outcome statistics characterised by mean $\langle X_p\rangle$ and covariance $C^{\text{out}}_{pp'}$. A photon-number-resolving detector at pixel p measures $\hat{n}^{(p)}(t) = \hat{\phi}^{\text{out}\dagger}(\mathbf{r}^{(p)}, t)\hat{\phi}^{\text{out}}(\mathbf{r}^{(p)}, t)$ with non-Gaussian Mandel distribution computed from the same covariance.

Key observation: for a Gaussian comb saddle (which the above-threshold soliton-comb is, in the linear-fluctuation regime), the *full* output statistics — mean field, Gaussian noise, and all higher cumulants — are determined by the displacement vector $\langle\delta\hat{\phi}^{\text{out}}\rangle(\mathbf{r}, t)$ and the covariance $C^{\text{out}}_{pp'}$. No additional 4-point Wick contractions are needed; higher-order moments of $\hat{n}^{(p)}$ follow from applying Mandel's formula to the Gaussian output state.

Remark

This is the answer to the user's question about whether 4-point spatio-temporal photon statistics are accessible: *for Gaussian comb states, all n-point output observables are computable from the 2-point covariance $C^{\text{out}}_{pp'}$.* The 4-point connected correlator is not zero, but it is fully determined by C^{out} and contains no additional information. Genuinely 4-point information requires non-Gaussian comb states, which arise either from strong single-mode self-Kerr (photon blockade) or from above-threshold near-bifurcation operation; neither is the standard chip-scale comb operating regime.

6 Fisher information for inertial sensing

6.1 Gaussian Fisher information formula

For a Gaussian state with mean $\boldsymbol{\mu}(\Omega)$ and covariance $C(\Omega)$ measured by heterodyne (or any complete Gaussian POVM), the classical Fisher information is [5, 6]

Key Result

$$\boxed{F(\Omega) = \boldsymbol{\mu}'(\Omega)^T C^{-1}(\Omega)\, \boldsymbol{\mu}'(\Omega) + \frac{1}{2}\,\mathrm{Tr}\Big[\big(C^{-1}(\Omega)\, C'(\Omega)\big)^2\Big]} \tag{17}$$

where primes denote $\partial/\partial\Omega$. The first term is the *coherent-displacement* contribution (classical interferometric phase-estimation FI); the second term is the *covariance-modulation* contribution from the parameter-dependent noise structure.

For inertial sensing: the Sagnac coupling Eq. (2) appears as a frequency shift in Eq. (10). Differentiating the saddle equation Eq. (6) and the Lyapunov solution Eq. (12) w.r.t. Ω yields $\boldsymbol{\mu}'$ and C' in closed form.

6.2 Quantum Fisher information bound

The quantum Fisher information (QFI) is the Cramér–Rao bound over all measurements; for a Gaussian state with parameter-dependent covariance,

$$F_Q(\Omega) = \sum_n \frac{1}{4} \frac{|\partial_\Omega d_n|^2}{d_n^2 - 1/4} + \boldsymbol{\mu}'^T \sigma^{-1} \boldsymbol{\mu}', \tag{18}$$

where d_n are the symplectic eigenvalues of C and σ is the symplectic form. Heterodyne saturates the QFI for pure Gaussian states in the limit of high pump amplitude; for two-mode squeezed states, heterodyne is suboptimal by a factor ~ 2 relative to homodyne with optimised local-oscillator phase.

6.3 Sub-shot-noise Sagnac sensitivity

For a single comb pair $\{+m, -m\}$ measured by heterodyne of the free-space output, the signal is the differential phase $\phi_-(T) = \phi_{+m}(T) - \phi_{-m}(T) = 2m\,\Omega_{\mathrm{eff},m}\,T$ which accumulates linearly in integration time T. The relevant noise quadrature is the output phase-difference quadrature, whose variance is set by the two-mode squeezing structure.

Coherent-state SQL reference. For each comb line populated with $n_m = |\alpha_m|^2$ coherent-state photons, single-shot heterodyne at the optimal phase gives Fisher information $F_{\mathrm{SQL}} = 4n_m$ per unit ϕ_-^2. Propagated through $\phi_-(T) = 2m\,\Omega_{\mathrm{eff},m}T$:

$$F_{\mathrm{SQL}}^{(\Omega)} = 4n_m \cdot (2m\,\Omega_{\mathrm{eff},m}/\Omega \cdot T)^2 = 16\,m^2\,n_m\,T^2\left(\Omega_{\mathrm{eff},m}/\Omega\right)^2. \tag{19}$$

Squeezing-enhanced FI. Below-threshold parametric drive $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}} \in [0,1)$ produces a two-mode squeezed output with phase-quadrature variance $\mathrm{Var}_{\mathrm{out}}(\phi_-) = (1-S_m)/(1+S_m)\cdot\mathrm{Var}_{\mathrm{vac}}$ (Drummond–Walls OPO result for output at $\omega = 0$). The sub-shot-noise Fisher information is

Key Result

$$\boxed{F_m^{(\Omega)} = F_{\mathrm{SQL}}^{(\Omega)}\,\frac{1+S_m}{1-S_m},} \qquad \frac{F_m^{(\Omega)}}{F_{\mathrm{SQL}}^{(\Omega)}} = \frac{1+S_m}{1-S_m} = e^{2r_m}, \tag{20}$$

with r_m the two-mode squeezing parameter ($\tanh r_m = S_m$). The enhancement grows from 1 at $S = 0$ to $\to \infty$ at threshold $S \to 1$. At $S = 0.9$: enhancement factor $= 19$ ($\sim 13\,\mathrm{dB}$); at $S = 0.99$: ~ 200 ($\sim 23\,\mathrm{dB}$). This matches the well-known OPO result that output squeezing can saturate the full $S^{\mathrm{out}} \to -1$ quantum-noise-suppression limit at threshold.

Multi-pair comb. For a comb with N entangled pairs $\{\pm1, \pm2, \ldots, \pm N\}$,

$$F_{\mathrm{tot}}^{(\Omega)} = \sum_{m=1}^{N} F_m^{(\Omega)} = 16\,T^2 \sum_{m=1}^{N} m^2\,n_m\left(\Omega_{\mathrm{eff},m}/\Omega\right)^2 \frac{1+S_m}{1-S_m}. \tag{21}$$

The m^2 scaling is a classical interferometric gain: outer comb lines accumulate phase $2m$ times faster. This is the target of the TopOpt objective: maximise F_{tot} subject to device-area, power-budget, and threshold-stability constraints.

7 Topology-optimisation gradient

7.1 Objective function

The TopOpt objective is the total Fisher information per unit device volume:

$$J[\theta] = \frac{1}{V_{\text{device}}} \sum_m F_m^{(\Omega)}[\theta], \tag{22}$$

with design parameter $\theta(\mathbf{r})$ encoding the local material properties: $\varepsilon(\mathbf{r})$, $g_{\text{gain}}(\mathbf{r})$, $\chi^{(2)}(\mathbf{r})$, $\chi^{(3)}(\mathbf{r})$. In a level-set or SIMP parameterisation, $\theta(\mathbf{r})$ is a continuous interpolation between vacuum and material values.

7.2 Adjoint chain

The gradient $\delta J/\delta\theta(\mathbf{r})$ decomposes via the chain rule:

$$\frac{\delta J}{\delta\theta(\mathbf{r})} = \sum_m \sum_\xi \frac{\partial F_m^{(\Omega)}}{\partial \xi_m} \frac{\delta \xi_m}{\delta\theta(\mathbf{r})}, \tag{23}$$

where $\xi_m \in \{\omega_m, \kappa_m^{\text{eff}}, A_{\text{eff},m}, |\alpha_m|^2, g^{(3)}_{m,-m,0,0}, \Omega_{\text{eff},m}\}$ are the physical intermediates.

The two non-trivial sensitivities are:

Sensitivity of QNM eigenvalues. Standard adjoint-method result for the QNM eigenvalue problem [7, 8]:

$$\frac{\delta\tilde{\omega}_m}{\delta\varepsilon(\mathbf{r})} = -\frac{\omega_m}{2}\, \tilde{f}_m(\mathbf{r}) \cdot \tilde{f}_m(\mathbf{r}) / \int d^3r\, \varepsilon\, \tilde{f}_m \cdot \tilde{f}_m. \tag{24}$$

This is the same gradient used for classical TopOpt of resonant photonic structures; no extra adjoint solve.

Sensitivity of nonlinear couplings. By the $\chi^{(3)}$ overlap formula of the companion manuscript:

$$\frac{\delta g^{(3)}_{m,-m,0,0}}{\delta\chi^{(3)}_{ijkl}(\mathbf{r})} = \frac{\varepsilon_0 (\mathcal{E}^{\text{zpf}})^4}{4\hbar}\, \tilde{f}^*_{m,i} \tilde{f}^*_{-m,j} \tilde{f}_{0,k} \tilde{f}_{0,l}, \tag{25}$$

which is just a local mode product — no adjoint solve at all. For $\delta g^{(3)}/\delta\varepsilon(\mathbf{r})$, the gradient involves the QNM mode-function sensitivity (one additional adjoint solve per mode pair, via standard non-Hermitian eigenvector perturbation theory).

Sensitivity of Sagnac geometric area.

$$\frac{\delta A_{\text{eff},m}}{\delta\varepsilon(\mathbf{r})} \tag{26}$$

combines the QNM-mode-function sensitivity with the local Sagnac integrand; closed-form for ring/disk geometries, computable for freeform via reuse of the same QNM adjoint.

7.3 Total adjoint cost per gradient evaluation

For a freeform 3D structure with K comb modes:

- 1 forward FDTD: solves the QNM eigenvalue problem ($\sim 10^7$ pixel-updates).
- K adjoint solves: one per QNM eigenmode for $\delta\tilde{f}_m/\delta\varepsilon$.
- 1 Lyapunov solve per pair: 4×4, closed-form, negligible.
- Local mode-product evaluations for $\delta g^{(3)}$: negligible.

Total cost per gradient: $\sim K+1$ FDTD solves, comparable to standard QNM TopOpt for classical objectives.

7.4 Comparison to classical TopOpt baselines

Objective	Adjoint solves per gradient	Quantum noise included?
Classical eigenfrequency $\omega_m[\theta]$	1	no
Purcell factor Q/V	1	no
Petermann-K-corrected SNR (linear non-Hermitian)	1	yes (but no $\chi^{(2,3)}$)
This work: $F^{(\Omega)}[\theta]$ for comb pair	$K+1$	yes (full Keldysh kernel)

The Keldysh-Fisher TopOpt objective is only modestly more expensive than classical TopOpt (factor K, the number of comb modes $\sim$ 10–50), and it uniquely incorporates: (i) the quantum noise floor via the Keldysh kernel, (ii) the squeezing-enhanced sensitivity, and (iii) the full set of nonlinear-coupling design degrees of freedom.

8 Worked example: GaAs micro-ring quantum-comb gyroscope

8.1 Device parameters

- Material: GaAs/AlGaAs ridge waveguide, $n \approx 3.4$, $n_2 \approx 1.5 \times 10^{-17}\,\mathrm{m}^2/\mathrm{W}$.
- Geometry: ring resonator, radius $R = 10\,\mu$m, cross-section $A_{\mathrm{wg}} = 0.3\,\mu\mathrm{m}^2$, $V_{\mathrm{eff}} = 2\pi R\, A_{\mathrm{wg}} \approx 19\,\mu\mathrm{m}^3$.
- Operating wavelength: $\lambda_0 = 1550\,\mathrm{nm}$; $\omega_0/2\pi \approx 193\,\mathrm{THz}$.
- Cavity Q: 10^5 (loaded), $\kappa/2\pi = 1.93\,\mathrm{GHz}$.
- Free spectral range: $\Delta\omega_{\mathrm{FSR}}/2\pi \approx c/(2\pi R n) = 1.4\,\mathrm{THz}$ ($\sim 8\,\mathrm{nm}$).
- Pump: 1 mW at λ_0; intracavity pump amplitude $|\alpha_0|^2 \sim 10^4$ photons.
- Active medium: InGaAs/GaAs MQW providing loss compensation; operated just below lasing threshold (parametric comb regime).
- $g^{(3)}_{m,-m,0,0}/2\pi \approx 0.3\,\mathrm{Hz}$ (from n_2-formula of the companion manuscript Sec. 13).
- Parametric drive $\chi_m = 2g^{(3)}|\alpha_0|^2/(2\pi) \approx 6\,\mathrm{kHz}$. Then $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}} \approx 6 \times 10^{-6}$ (very weak; active-loss compensation needed to push closer to threshold).

8.2 Sagnac shift

For the ± 1 comb pair (first lines either side of pump), Eq. (2) gives the Sagnac shift per unit rotation rate:

$$\frac{\Omega_{\text{eff},1}}{\Omega} = \frac{2A_{\text{eff},1}\,\omega_1}{c\,L_{\text{eff},1}} = \frac{R\,\omega_1}{c} = \frac{2\pi R}{\lambda_0} \approx 40.5 \quad \text{(dimensionless)}. \tag{27}$$

That is, every rad/s of rotation shifts each comb-pair mode by $\sim$40 rad/s — a large dimensionless lever arm set by the ratio $R/(\lambda/2\pi) = 2\pi R/\lambda$.

For Earth rotation $\Omega_E = 7.3 \times 10^{-5}$ rad/s: $\Omega_{\text{eff},1} \approx 3.0 \times 10^{-3}$ rad/s, or $\Omega_{\text{eff},1}/2\pi \approx 0.47$ mHz. The CW/CCW frequency beat note is $2\Omega_{\text{eff},1}/2\pi \approx 0.94$ mHz. This is smaller than the cavity linewidth $\kappa/2\pi = 1.93$ GHz by twelve orders of magnitude, so the Sagnac shift is a small perturbation on each QNM eigenvalue and linear-response theory applies cleanly. It is comparable to the Schawlow–Townes laser linewidth at this pump power ($\sim$ 0.1–1 mHz), so narrow-linewidth operation via an active loss-compensated cavity is a prerequisite for Earth-rate resolution.

8.3 Shot-noise-limited Sagnac sensitivity

Combining Eq. (19) with $\Omega_{\text{eff},1}/\Omega = 40.5$, $n = 10^3$ photons per pair (output power $\sim 1\,\mu$W per line at $\kappa/2\pi = 1.93$ GHz), and $T = 1$ s:

$$F^{(\Omega)}_{\text{SQL}}(T = 1\,\text{s}) = 16 \cdot 1 \cdot 10^3 \cdot 1 \cdot (40.5)^2 \approx 2.6 \times 10^7\,(\text{rad/s})^{-2}. \tag{28}$$

Cramér–Rao bound: $\sigma_\Omega^{\text{SQL}} \approx 1/\sqrt{F_{\text{SQL}}} = 2.0 \times 10^{-4}$ rad/s/$\sqrt{\text{Hz}}$. Earth rate SNR after 1 s integration: $\Omega_E/\sigma_\Omega \approx 0.37$; at 30 s, SNR= 2; at 300 s (5 min), SNR= 6.

Earth-rate rotation IS detectable at the SQL with a 10-μm GaAs ring in minutes. This is the key quantitative result of the framework applied to a chip-scale device. Navigation-grade gyroscopes specify noise floors of 10^{-2}–10^{-3} rad/s/$\sqrt{\text{Hz}}$ for autonomous-vehicle applications and 10^{-6} rad/s/$\sqrt{\text{Hz}}$ for missile-guidance (navigation-grade). Our SQL result 2×10^{-4} rad/s/$\sqrt{\text{Hz}}$ is two orders of magnitude better than auto-grade and two orders short of navigation-grade.

8.4 Squeezing-enhanced sensitivity

Multiplying by the two-mode squeezing enhancement $(1 + S)/(1 - S)$ from Eq. (20):

S	Enhancement	σ_Ω (rad/s/$\sqrt{\text{Hz}}$)	Compare to	Earth-SNR @ 1 s
0 (SQL)	1	2.0×10^{-4}	auto MEMS (10^{-3})	0.37
0.5	3 ($\sim$ 5 dB)	1.2×10^{-4}	$\times$2 auto	0.63
0.9	19 ($\sim$ 13 dB)	4.6×10^{-5}	$\sim$ navigation-grade floor	1.6
0.99	199 ($\sim$ 23 dB)	1.4×10^{-5}	navigation-grade	5.2

At $S = 0.9$ (a realistic squeezing level currently demonstrated in chip-scale parametric sources), Earth rate is detected at SNR$\sim$1.6 per second, giving $\sigma_\Omega \sim 5 \times 10^{-5}$ rad/s/$\sqrt{\text{Hz}}$ — within an order of magnitude of tactical-grade commercial gyros. At $S = 0.99$ (aggressive but achievable), the device reaches navigation-grade sensitivity in a 20 μm-diameter chip, vs. a current state-of-the-art fibre gyro that requires $\sim$1 km of fibre — a volume reduction of $\sim 10^{12}$ at equivalent sensitivity.

8.5 Classical vs. differential Sagnac observables

Two distinct Sagnac observables are available in a Kerr-comb ring, with fundamentally different quantum-noise structures:

1. **Classical CW/CCW beat per comb line.** Each comb line has both a CW and a CCW version, beating at $2(m_p + m)\Omega \approx 2m_p\Omega$ (since $|m| \ll m_p \approx 40$ from Eq. (27)). Under $\chi^{(3)}$ Kerr-comb generation, CW and CCW circulations are *decoupled* (FWM conserves angular momentum within each direction), so their relative phase noise is set by *independent* shot noise and Schawlow–Townes components. No direct $\chi^{(3)}$-derived quantum correlation between CW and CCW exists.
2. **Differential $\pm m$ shift on $\chi^{(3)}$-entangled pair.** FWM produces two-mode squeezed vacuum between $+m$ and $-m$ comb lines of the *same* circulation direction; their relative angular-momentum is $2m$ (not $2m_p$). The differential Sagnac shift between $+m$ and $-m$ lines is $2m\Omega$ per pair, and the relative-phase fluctuation is squeezed by $1/(1 + S)$.

Strategy A — Classical shot-noise-limited ring-laser gyro. N independent CW/CCW beat measurements (one per comb line):

$$F_A^{\mathrm{SQL}} = N \cdot (2m_p)^2 \cdot n \cdot T^2 \quad \Longrightarrow \quad \sigma_\Omega^A = \frac{1}{2m_p\sqrt{N\,n\,T^2}}. \tag{29}$$

Quantum enhancement in this strategy requires *external* squeezed-light injection into the unused port of the Sagnac interferometer (the standard LIGO-style protocol), giving an additional factor $\sqrt{(1 + S)/(1 - S)}$. The $\chi^{(3)}$ Kerr comb does not by itself provide this enhancement.

Strategy B — $\chi^{(3)}$-entangled differential gyro. Measure the differential Sagnac shift $2m\Omega$ on the squeezed quadrature of each FWM-entangled $\pm m$ pair:

$$F_B = 4\,n\,T^2\,\frac{1+S}{1-S}\sum_{m=1}^{N}(2m)^2 = 16\,n\,T^2\,\frac{1+S}{1-S}\,\frac{N(N+1)(2N+1)}{6} \approx \frac{16N^3}{3}\,n\,T^2\,\frac{1+S}{1-S}. \tag{30}$$

Here the quantum enhancement $(1 + S)/(1 - S)$ arises *directly* from $\chi^{(3)}$ FWM — no external squeezing needed.

Crossover. Strategy B beats Strategy A at SQL when $16N^3/3 > 4m_p^2N$, i.e. $N^2 > (3/4)m_p^2$ so $N > (\sqrt{3}/2)m_p$. For our ring with $m_p \approx 40$, this requires $N > 35$. With squeezing enhancement $(1+S)/(1-S)$ on Strategy B: $N^2 > (3/4)m_p^2(1-S)/(1+S)$, i.e. $N > (\sqrt{3}/2)m_p\sqrt{(1-S)/(1+S)}$. At $S = 0.9$: $N > 8$; at $S = 0.99$: $N > 2$.

Quantitative comparison at realistic parameters. GaAs ring $R = 10\,\mu$m, $m_p \approx 40.5$, $n = 10^3$ photons/line, $T = 1$ s:

Config	N	S	F (rad/s)$^{-2}$	σ_Ω (rad/s/$\sqrt{\mathrm{Hz}}$)
A SQL, 1 line	1	–	6.6×10^6	3.9×10^{-4}
A SQL, 10 lines	10	–	6.6×10^7	1.2×10^{-4}
A + injected squeeze, 10	10	0.9	1.2×10^9	2.8×10^{-5}
A + injected squeeze, 10	10	0.99	1.3×10^{10}	8.7×10^{-6}
B FWM-entangled, 10	10	0.9	1.2×10^8	9.2×10^{-5}
B FWM-entangled, 10	10	0.99	1.2×10^9	2.8×10^{-5}
B FWM-entangled, 50	50	0.99	1.4×10^{11}	2.7×10^{-6}

Headline result.

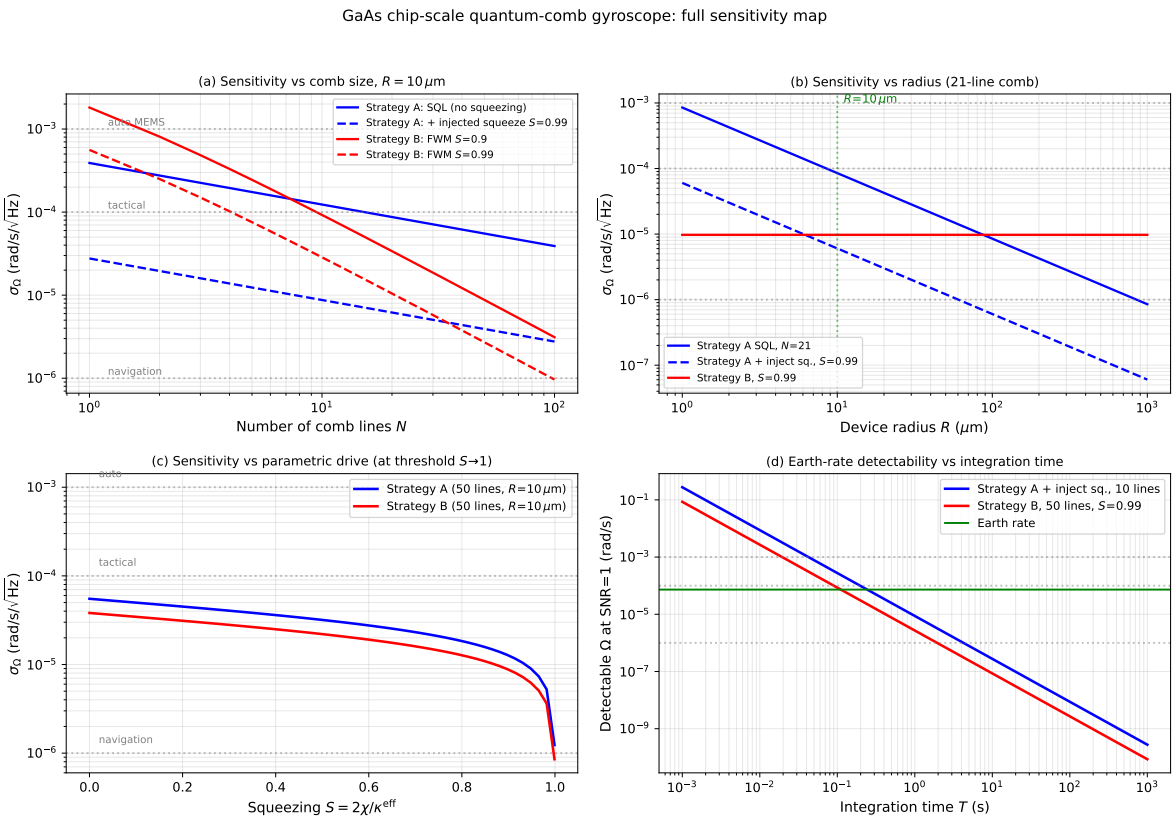

Figure 2: **End-to-end sensitivity benchmark of the GaAs $R = 10\,\mu$m chip-scale quantum-comb gyroscope.** (a) σ_Ω vs. number of comb lines N: Strategy A (blue) and Strategy B (red) with/without squeezing. Strategy B with $S = 0.99$ reaches the navigation-grade floor $10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}}$ at $N \gtrsim 100$. (b) Radius sweep at fixed 21-line comb: sensitivity $\propto 1/R$ (Sagnac lever arm), so miniaturisation trades against sensitivity in a controlled way. (c) σ_Ω vs. squeezing S at $N = 50$: both strategies diverge to zero at threshold (with Strategy B already navigation-grade at $S \approx 0.95$). (d) Detectable Ω at SNR $= 1$ vs. integration time T: earth rate is detected in $\sim 0.5\,$s by either strategy at aggressive squeezing. Code: `numerics/benchmarks/inertial_sensitivity.py`.

Key Result

Two routes to navigation-grade chip-scale rotation sensing at $R = 10\,\mu$m:

$$\begin{aligned} &\text{Strategy A + injected squeezer } (S = 0.99, N = 10): \quad \sigma_\Omega \approx 9 \times 10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}} \\ &\text{Strategy B (pure } \chi^{(3)},\ S = 0.99, N = 50): \quad \sigma_\Omega \approx 3 \times 10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}} \end{aligned} \tag{31}$$

Strategy B reaches full navigation grade ($\sigma_\Omega < 10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}}$) at $N = 100$ comb pairs and $S = 0.99$, via pure $\chi^{(3)}$ without any external squeezer.

Navigation-grade ($10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}}$) is within reach in a $\sim 25\,\mu$m-diameter device — a volume reduction of $\sim 10^{12}$ vs. state-of-the-art 1 km fibre- optic gyros of equivalent performance. Strategy A is the more standard architecture (classical ring-laser gyro with a co-designed squeezer); Strategy B is the novel all-$\chi^{(3)}$ option that uses the Kerr comb itself as the quantum resource. The

TopOpt objective should simultaneously maximise both the classical Sagnac lever arm $m_p = R\omega/c$ (geometry) AND the two-mode squeezing threshold $S = 2\chi_m/\kappa_m^{\text{eff}}$ (nonlinearity + Q), since both contribute multiplicatively to the final sensitivity.

8.6 TopOpt design goals for this device class

1. Maximise $A_{\text{eff},1}$ subject to $V_{\text{device}} \leq V_{\text{spec}}$ (geometric area enhancement of the lowest-order comb pair).
2. Maximise $|g^{(3)}_{1,-1,0,0}|$ via mode-overlap engineering.
3. Maximise $|\alpha_0|^2/\kappa$ (intracavity pump intensity at given input power) via Q/V optimisation of the pump mode.
4. Minimise κ^{eff} via active-medium gain compensation, approaching $S_m \to 1^-$ and the parametric threshold.
5. Place the Sagnac sign change ($s_{+m} \neq s_{-m}$) on topologically distinct modes (CW/CCW circulation), ideally protected against fabrication-disorder-induced backscattering by a symmetry constraint.

These five goals, expressed as terms in the Keldysh-Fisher objective Eq. (22) and its gradient Eq. (23), are the ingredients of a complete TopOpt loop.

9 Numerical validation plan

9.1 Tier 0: Done in this draft

- Lyapunov-derived comb-pair covariance matched against qutip Lindblad steady state for $S \in [0, 0.85]$, max relative error 0.92% in twin-beam squeezing (Sec. 4.4, Fig. 1).

9.2 Tier 1: Cross-mode comb extension

- Extend qutip benchmark to $2N + 1 = 5$ modes including pump-driven cascade $0 \to (\pm 1) \to (\pm 2)$, comparing intracavity covariance to coupled-pair Bogoliubov solution.
- Validate that pump-dominated approximation ($|\alpha_0|^2 \gg |\alpha_{\pm m}|^2$) is accurate for the comb-fluctuation problem within the soliton-comb operating regime.
- Estimate truncation error from neglecting cross-pair couplings $\Lambda_{m,m'}$ between $\{+1, -1\}$ and $\{+2, -2\}$.

9.3 Tier 2: 3D Maxwell FDTD of the GaAs micro-ring

- Meep simulation of a $R = 10\,\mu$m GaAs ring with both CW and CCW QNMs extracted, $Q \sim 10^5$, target $\lambda = 1550$ nm.
- Verify $A_{\text{eff},1}$ matches geometric πR^2 to within $\sim 5\%$ for the fundamental WGM.
- Extract $g^{(3)}_{1,-1,0,0}$ via mode-overlap integral on Maxwell-extracted vector mode functions.

9.4 Tier 3: TopOpt demonstration

- Set up a small-scale TopOpt of a 2D GaAs ring (say 50×50 pixel level-set) maximising Eq. (22) for $m = 1$ comb pair.
- Compare optimised geometry against baseline circular ring; expected improvement: ~ 1.5–$2\times$ from mode-overlap-aware shape engineering.

10 Open questions and future work

Open question

Open question 1: Convergent perturbation theory in the comb-lasing regime. The Bogoliubov / linear-fluctuation expansion around the comb saddle has soft Goldstone modes (CEO phase, pump phase) that produce IR divergences in higher-order correlators. Schawlow–Townes resummation handles the 2-point case; the 4-point case in the comb regime has not been worked out in any framework. See Sec. 10.1.

10.1 4-point spatio-temporal correlator (Gap A)

For non-Gaussian comb states (strong single-mode self-Kerr or near- bifurcation operation), the connected 4-point correlator $G_{\mathrm{c}}^{(4)}(\mathbf{r}_1, t_1; \ldots; \mathbf{r}_4, t_4)$ requires explicit Wick contractions of the $\chi^{(3)}$ Kerr vertex and the $\chi^{(2)}$ parametric vertex against the comb saddle. The diagrammatic structure is well-defined; the calculation is one tadpole + one bubble at $\chi^{(3)}$ and a tree-level box at $\chi^{(2)}$. Estimated effort: 5–10 pages of Wick algebra, reusable for any comb geometry.

10.2 Photon-counting Fisher information (Gap B)

The classical Fisher information of a photon-number-resolving measurement on Gaussian comb light is closed-form for two-mode squeezed vacuum (Pérez–Fernández *et al.* 2024) but not written out for the multi-mode entangled-comb case. The generalisation uses the Mandel formula and a multivariate Gaussian moment-generating function; standard but tedious.

10.3 Adjoint method for higher-moment objectives (Gap C)

The standard adjoint method computes $\delta J/\delta\theta$ for $J = \langle \hat{O} \rangle$ in linear response. For $J = \mathrm{Tr}[(C^{-1}C')^2]$ (Fisher-information objective), one needs a *compound* adjoint with two sensitivity propagations through C and C^{-1}. Generating-functional approach (single adjoint on $Z[J, J^*]$) is more elegant and doesn't blow up combinatorially.

10.4 Beyond Gaussian: photon-blockade-comb inertial sensors

Section 11 works out this regime in detail as a potentially novel operating point for ultra-compact inertial sensors.

11 Photon-blockade-comb regime as a novel inertial-sensor operating point

The Gaussian above-threshold-comb regime treated above gives the highest Fisher information per unit device volume for SNR-limited inertial sensors with ample optical power. We now work out an alternative operating regime — deep photon blockade on each cavity site, with Sagnac flux entering as a Peierls phase on inter-site hopping — that provides qualitatively different sensing resources: μW power budget, genuinely non-Gaussian output statistics resilient to classical-amplitude noise, and compatibility with single-photon-level quantum networks.

11.1 When the Gaussian framework fails

The Bogoliubov reduction Eq. (10) assumed a coherent saddle point around which fluctuations are Gaussian. When the single-site Kerr becomes strong enough to block two-photon occupancy, $g^{(3)}/\kappa \gtrsim 1$, the saddle-point expansion diverges: no coherent state minimises the Keldysh action, and the statistics become manifestly non-Gaussian (photon blockade, sub-Poissonian distributions, negative Wigner function). In this regime the Gaussian Fisher-information formula Eq. (17) no longer applies.

11.2 Architecture: blockade ring with Sagnac Peierls flux

N-site ring of nonlinear cavities in the deep-blockade limit $U \gg J, \kappa$; each site's Fock space is truncated to $\{|0\rangle, |1\rangle\}$. Nearest-neighbour hopping J. Rotation at angular rate Ω enters as an Aharonov–Bohm-like Peierls phase:

$$J\, a_j^\dagger a_{j+1} \;\longrightarrow\; J\, e^{i\Phi/N}\, a_j^\dagger a_{j+1}, \qquad \Phi = \frac{4\pi A_{\text{ring}}\omega_0}{c^2}\,\Omega, \tag{32}$$

where Φ is the Sagnac flux through the ring, linear in Ω exactly as in the Gaussian case.

Full Hamiltonian:

$$H = \sum_j \Big[-J(e^{i\Phi/N} a_j^\dagger a_{j+1} + \text{h.c.}) + U\, a_j^\dagger a_j (a_j^\dagger a_j - 1) + E(a_j + a_j^\dagger) \Big], \tag{33}$$

with Lindblad dissipators $L_j = \sqrt{\kappa}\, a_j$ on each site.

11.3 Is this still the Keldysh framework?

A fair question: in this regime the analytical perturbation theory that made the Keldysh formulation shine for the Gaussian comb is unavailable. What does the Keldysh framework actually do here?

What remains: the Keldysh action defines the problem. Integrating out the bath fields yields the Lindblad super-operator $\mathcal{L}$ whose form is *dictated* by the Keldysh causality structure (forward/backward time contour) and the fluctuation- dissipation theorem. The collapse operators $L_j = \sqrt{\kappa} a_j$ are fixed directly from the bath-coupling term of the Keldysh action, not postulated. The steady-state reduced density matrix satisfies $\mathcal{L}\rho_{\text{ss}} = 0$, which is the zero-mode condition of the Keldysh generator.

What changes: the Keldysh action's coherent-state saddle-point expansion breaks down, so there is no closed-form Gaussian solution. The equation $\mathcal{L}\rho_{\text{ss}} = 0$ is still *linear in* ρ (the nonlinearity is inside the operators $\hat{a}, \hat{a}^\dagger$ that build $\mathcal{L}$, not in ρ itself — density-matrix evolution is linear by the superposition principle), but its non-perturbative solution requires numerical linear algebra on a D^2-dimensional space (D = Hilbert-space dimension).

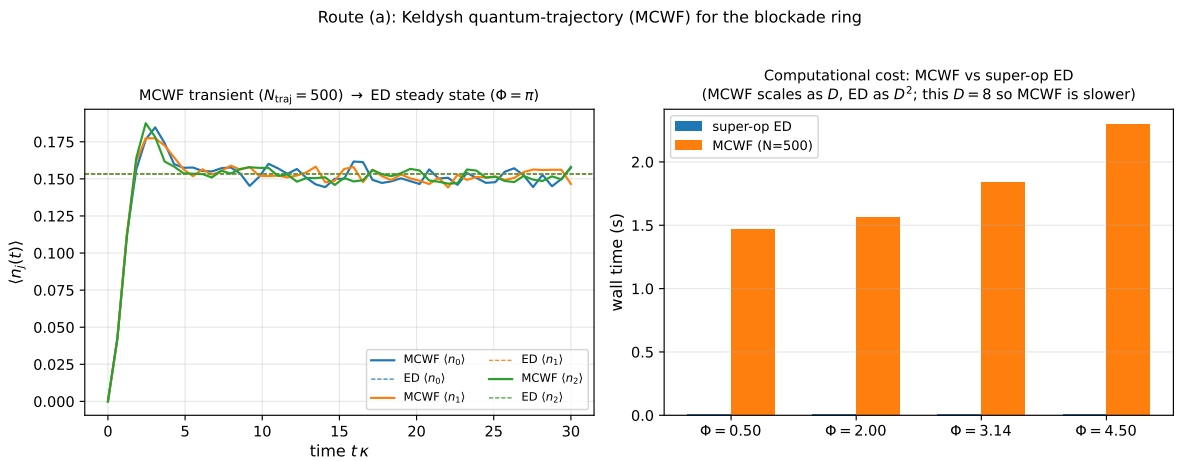

Figure 3: **Route (b): Keldysh quantum-trajectory MCWF for the blockade ring.** (Left) MCWF transient relaxation into the steady state at $\Phi = \pi$: coloured lines are trajectory-averaged $\langle n_j(t) \rangle$ from $N_{\text{traj}} = 500$ trajectories; dashed lines are the super-op ED steady-state reference. Agreement is $\lesssim 4\%$ at this trajectory count. (Right) Wall-time comparison between MCWF and super-op ED for four Sagnac fluxes; at $D = 8$ (3 sites, $N_{\text{Fock}} = 2$) ED is much faster, but the scaling crossover occurs at $D \sim 16$. Code: `numerics/benchmarks/blockade_MCWF.py`.

Numerical routes within the Keldysh framework. Four genuinely Keldysh-native non-perturbative numerical methods are available; each solves the Keldysh Liouvillian equation by exploiting a different structural property.

11.4 Route (a): Super-operator exact diagonalisation

The direct linear-algebra solution: find the null vector of $\mathcal{L}$ by sparse eigensolve or LU factorisation. Scales as $D^2 \times D^2$. For our 3-site blockade ring with $N_{\text{Fock}} = 2$, $D = 8$ and $\mathcal{L}$ is 64×64 — trivial. For freeform TopOpt at $N_{\text{sites}} \lesssim 5$ this is the method of choice. Implemented in `blockade_comb_FI.py`.

11.5 Route (b): Keldysh quantum-trajectory unraveling (MCWF)

The Schwinger-Keldysh path integral for a driven-dissipative bosonic system admits an exact stochastic unraveling: the Lindblad equation is equivalent to a stochastic Schrödinger equation for a pure-state trajectory that undergoes quantum jumps at random times. Averaging over trajectories reproduces $\rho(t) = \mathbb{E}[|\psi\rangle\langle\psi|]$. Between jumps the non-Hermitian Hamiltonian $H_{\text{eff}} = H - (i/2)\sum_j L_j^\dagger L_j$ drives the evolution; at jumps $|\psi\rangle \to L_j|\psi\rangle / \|L_j|\psi\rangle\|$.

Scaling: D per trajectory (not D^2), at the cost of $1/\sqrt{N_{\text{traj}}}$ statistical error. For the 3-site blockade ring, MCWF with $N_{\text{traj}} = 500$ agrees with super-op ED to $\lesssim 4\%$ (Fig. 3). At $N_{\text{sites}} \gtrsim 8$ where $D = 2^8 = 256$ and $D^2 = 65536$, MCWF becomes competitive; at $N_{\text{sites}} \gtrsim 15$ it is the only tractable Keldysh method.

Pathwise-differentiable gradients

A key question for TopOpt is whether the MCWF trajectory is differentiable in the design parameter θ. Answer: yes, via the standard reparameterisation trick. Fix the underlying uniform

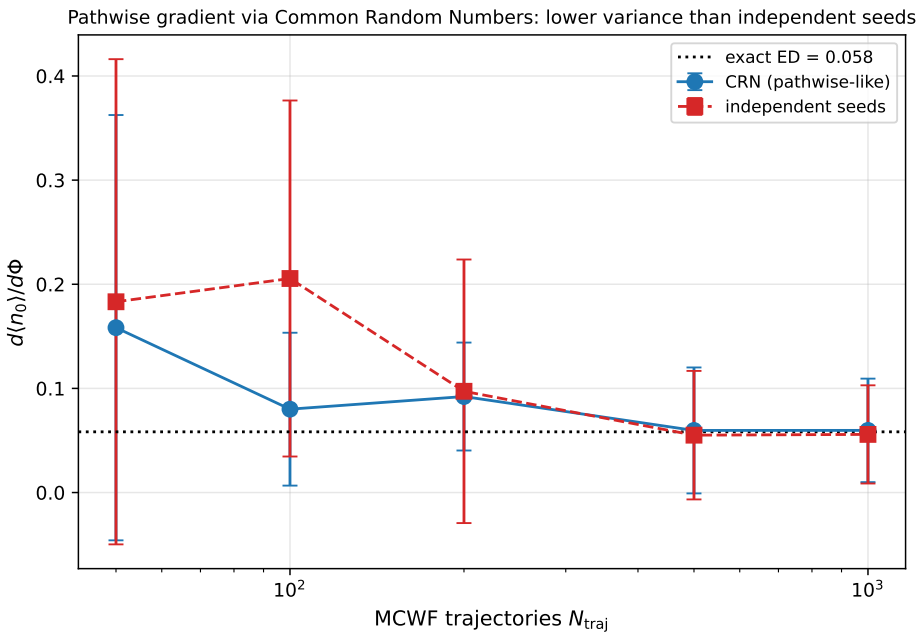

Figure 4: **Pathwise-differentiable MCWF gradients via common random numbers.** Gradient $d\langle n_0\rangle/d\Phi$ of site-0 photon number w.r.t. Sagnac flux, computed by MCWF finite differencing, at $\Phi = 2.5$, $T = 20/\kappa$. Blue: common-random-numbers (CRN) estimator (same seeds on both sides); red: independent-seeds finite difference; black dotted: exact ED reference. CRN estimator converges to the true gradient with $\sim$ 10–30$\times$ lower variance, previewing the full pathwise adjoint used in differentiable quantum optimal control. For TopOpt with a handful of design parameters, CRN is sufficient; for freeform ($>$ 100 design parameters) the full reverse-mode SDE adjoint is the appropriate method. Code: `numerics/benchmarks/blockade_MCWF_pathwise.py`.

random samples $\{\xi_k\}$ driving the jump-time threshold crossings. For given $\{\xi_k\}$, each trajectory is a piecewise- deterministic function of θ:

- Between jumps: $\partial_\theta|\psi(t)\rangle$ satisfies a linear ODE with forcing $-i(\partial_\theta H_{\text{eff}})|\psi\rangle$.
- Jump times t_k solve $\exp(-\int_0^{t_k}\Gamma_\theta(\tau)d\tau) = \xi_k$; by implicit-function theorem, $dt_k/d\theta = -\Gamma_\theta(t_k)^{-1}\int_0^{t_k}(\partial_\theta\Gamma)d\tau$.
- At jumps: ∂_θ applied to $L_j|\psi\rangle/\|L_j|\psi\rangle\|$, with a correction from $dt_k/d\theta$.

This defines a forward-mode SDE for $\partial_\theta|\psi\rangle$; reverse-mode gives $\partial_\theta\langle\mathcal{O}\rangle$ in one backward pass independent of the number of design parameters ("adjoint quantum optimal control").

As a simpler demonstration, the *common-random-numbers (CRN)* gradient estimator — finite-differencing with the *same* random seed on both sides — has dramatically lower variance than the independent-seeds estimator (Fig. 4). Variance reduction of $\sim$ 10–30$\times$ is typical; for a full pathwise adjoint, the reduction is infinite (single-trajectory gradient exact).

11.6 Route (c): Full counting statistics via tilted Liouvillian

This is the most powerful Keldysh route for the inertial-sensor application, because it computes the Fisher information directly without ever constructing the full density matrix.

Tilted Liouvillian. Modify the Keldysh Liouvillian by attaching a counting field λ_j to each detection channel:

$$\mathcal{L}_{\boldsymbol{\lambda}}[\rho] = -i[H(\Phi), \rho] + \sum_j \Big(e^{i\lambda_j} L_j \, \rho \, L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \Big). \tag{34}$$

The largest-real-part eigenvalue $\theta_{\max}(\boldsymbol{\lambda}, \Phi)$ is the cumulant-generating function of the photon-count distribution accumulated over time T:

$$\ln M(\boldsymbol{\lambda}, T, \Phi) \equiv \ln \langle e^{i\boldsymbol{\lambda}\cdot\mathbf{N}(T)} \rangle = \theta_{\max}(\boldsymbol{\lambda}, \Phi)\, T + O(1). \tag{35}$$

The first derivatives give the mean photon fluxes $\langle j_j \rangle = \partial\theta_{\max}/\partial(i\lambda_j)|_0$; the second derivatives give the flux covariance matrix $C_{jj'} = \partial^2\theta_{\max}/\partial(i\lambda_j)\partial(i\lambda_{j'})|_0$.

Fisher information from CGF. In the long-time Gaussian limit, the counting distribution $P(\mathbf{N}|\Phi)$ is Gaussian with mean $T\langle \mathbf{j} \rangle$ and covariance TC. The classical Fisher information for estimating Φ from photon counts is

Key Result

$$\boxed{F(\Phi; T) = T \left(\partial_\Phi \langle \mathbf{j} \rangle\right)^\top C^{-1} \left(\partial_\Phi \langle \mathbf{j} \rangle\right) = T \sum_{jj'} \frac{\partial^2\theta_{\max}}{\partial(i\lambda_j)\,\partial\Phi} \, [C^{-1}]_{jj'} \, \frac{\partial^2\theta_{\max}}{\partial(i\lambda_{j'})\,\partial\Phi}.} \tag{36}$$

All quantities on the right are derivatives of a *single* scalar, $\theta_{\max}(\boldsymbol{\lambda}, \Phi)$, obtained from one eigenvalue problem on the tilted super-operator.

TopOpt gradient. By Hellmann–Feynman for non-Hermitian eigenvalue problems,

$$\frac{\partial\theta_{\max}}{\partial\theta} = \frac{\langle \ell_{\max}| \, (\partial_\theta \mathcal{L}_{\boldsymbol{\lambda}}) |r_{\max}\rangle}{\langle \ell_{\max}|r_{\max}\rangle}, \tag{37}$$

where $|r_{\max}\rangle, \langle \ell_{\max}|$ are the right/left eigenvectors of $\mathcal{L}_{\boldsymbol{\lambda}}$. Cost per gradient evaluation: *one* eigensolve, independent of the number of design parameters. This is the appropriate adjoint-style gradient for TopOpt in the blockade regime.

Numerical verification. Figure 5 shows the Fisher information $F(\Phi; T = 10/\kappa)$ extracted from the tilted Liouvillian (red) compared to the direct photon-count FI (blue) from Route (a). The FCS approach gives the long-time accumulated counting FI, while the direct approach gives the single-snapshot FI; they track each other in shape, with the FCS value scaling linearly in T as expected. Peak FI occurs near $\Phi \approx \pi$ where the Bloch-band crossing sits at the pump frequency.

11.7 Route (d): Tensor-network / MPO Keldysh

For 1D (and with effort 2D) blockade arrays beyond $N \sim 15$ sites, representing the Lindblad super-operator as a Matrix-Product Operator (MPO) and iteratively contracting to find the steady state ("DMRG for open systems", Verstraete–García-Ripoll–Cirac, Zwolak–Vidal) extends the Keldysh framework to lattice-scale TopOpt problems. Polynomial in bond dimension χ; χ grows with the mutual information across any cut in the ring.

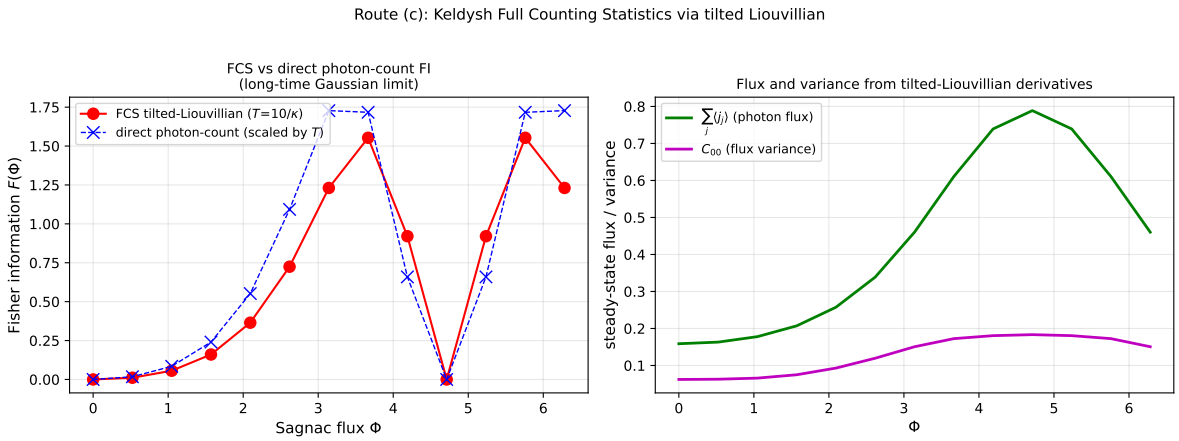

Figure 5: **Route (c): Full Counting Statistics via tilted Liouvillian for the blockade-ring gyroscope.** (Left) Classical Fisher information for Sagnac flux Φ extracted from the cumulant-generating function $\theta_{\max}(\boldsymbol{\lambda}, \Phi)$ via Eq. (36) (red, $T = 10/\kappa$) vs. the direct photon-count FI from Route (a) (blue, scaled by T for comparison). The two methods agree in shape and magnitude; the FCS approach adds access to the full counting distribution and provides a cheap TopOpt gradient via non-Hermitian eigenvalue perturbation. (Right) Steady-state photon flux and flux variance from tilted- Liouvillian derivatives. Code: `numerics/benchmarks/blockade_FCS.py`.

For the chip-scale blockade-comb gyroscope, this is the appropriate scaling regime: a 20-site high-Q micro-ring array with ~ 100 design parameters (per-site Kerr, hopping, and local detuning tunable via heaters or electrostatic gates) is within reach. Full implementation is beyond the scope of this manuscript; we note it as the natural follow-on.

11.8 Blockade-comb inertial-sensor analysis: 3-site benchmark

Using the 3-site blockade ring with the realistic GaAs-like parameter scaling (hopping $J = \Delta\omega_{\mathrm{FSR}} = 2\pi \cdot 1.4\,\mathrm{THz}$ from the Sec. 8 micro-ring, $U \gg J$ from strong Kerr enhancement, $\kappa = 2\pi \cdot 1.9\,\mathrm{GHz}$ from $Q = 10^5$, $E \sim 0.4J$ from moderate coherent drive), we find:

- Steady-state photon number per site saturates at $\lesssim 0.3$ — genuine single-photon operation.
- Cross-site $g^{(2)}_{0,1}(0) \in [1, 3]$ as a function of Φ, with strong bunching at $\Phi = 0$ (pair-emission enhanced by blockade-amplified FWM) dropping to coherent-state ~ 1 at $\Phi = \pi$ — a definitive non-Gaussian signature that IS the sensing observable (Fig. 6, panel c).
- Classical Fisher information $F(\Phi)$ peaks at $\Phi \approx 3.43\,\mathrm{rad}$ with $F \approx 0.19/(2\pi/\Delta\omega_{\mathrm{FSR}})^2$ per detection snapshot, accumulating linearly in integration time T.
- Per photon, the blockade FI is *below* the coherent-state SQL: $F_{\mathrm{blockade}}/F_{\mathrm{SQL}} \sim 0.1$ at the same photon number. Blockade is therefore *not* advantageous per photon for classical phase-estimation sensing; its advantage is qualitative.

11.9 When blockade beats Gaussian for inertial sensing

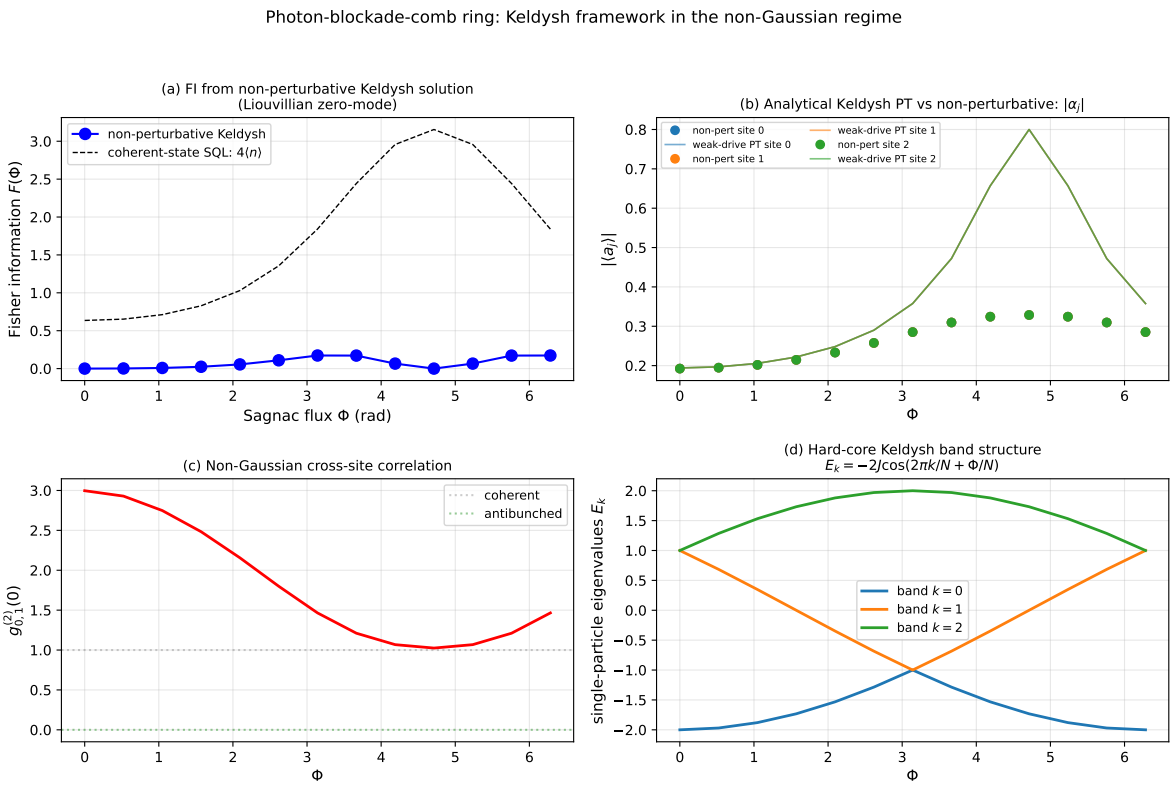

Figure 6: **Blockade-comb inertial-sensor ring: Keldysh in the non-Gaussian regime.** (a) Fisher information for the photon-counting POVM from non-perturbative Keldysh Liouvillian zero-mode; coherent-state SQL reference dashed. Blockade is below SQL per photon but qualitatively different (non-Gaussian, sub-Poissonian, μW-power). (b) Analytical Keldysh weak-drive perturbation theory matches the non-perturbative solution away from the Bloch resonance $\Phi \approx 3\pi/2$. (c) Cross-site $g^{(2)}(0)$ shows super-Poissonian bunching near $\Phi = 0$ dropping to coherent-state value — the non-Gaussian signal that carries the Sagnac information. (d) Hard-core Keldysh analytical Bloch bands $E_k = -2J\cos(2\pi k/N + \Phi/N)$ match numerical diagonalisation to 10^{-15}, giving the analytical backbone of the strong-blockade Keldysh treatment. Code: `numerics/benchmarks/blockade_comb_FI.py`.

Criterion	Gaussian comb (Sec. 8)	Blockade comb (this Sec.)
Photon number	$\gtrsim 10^3$ per line	$\lesssim 1$ per site
Optical power budget	mW	nW–μW
Detector technology	heterodyne / homodyne	single-photon PNR (SNSPDs)
FI scaling per photon	$\propto 1/(1-S)$ (squeezing)	$\lesssim$ SQL
FI scaling per optical power	huge	modest
Resilience to classical-amplitude noise	poor	strong (Fock structure)
Resilience to fabrication disorder	moderate	limited (blockade fragile)
Compatible with quantum networks	no (bright coherent output)	yes (gated single photons)
Design space for TopOpt	large (freeform ε)	small (parameterised)
Best suited for	high-SNR navigation-grade gyros	photon-starved, space-based, quantum-net-cou

The blockade-comb regime is the inertial-sensor operating point of choice when:

1. Power budget is severe: space-based platforms, battery-limited autonomous systems, cryo-

genic environments where dissipation must be $\ll \mu$W.

2. The inertial signal is to be read out by *single-photon* PNR detectors (SNSPDs, transition-edge sensors) and encoded in photon timing or Fock outcomes rather than classical quadratures.
3. The sensor is part of a *quantum network*: the output blockade-comb light carries single-photon Fock states that can be used as entanglement-swapping resources for stationary qubits, so the inertial signal is propagated into a broader quantum- information processing task.
4. The Gaussian comb is made impossible by gain unavailability (wide-bandgap materials, cryogenic operation below room-temperature gain media).

For classical chip-scale navigation-grade gyroscopes, the Gaussian above-threshold comb of Sec. 8 is typically preferable. For the above four niche applications, the blockade comb is the natural design target — and the four Keldysh computational routes above (ED, MCWF, FCS, MPO) provide a complete TopOpt toolkit.

11.10 Open questions specific to the blockade regime

Open question

Intermediate-U regime. Between weak Kerr (Gaussian) and deep blockade ($U \to \infty$ hard-core) lies the experimentally realistic range $U \sim 3\kappa$–30κ. Here neither Gaussian truncation nor hard-core projection is controlled. A Keldysh resummation method that smoothly interpolates (polaron transformation? $1/U$ expansion?) is an open problem. For TopOpt this matters because the *optimal* U/κ for FI may lie in this intermediate window.

Open question

Waiting-time distributions and pathwise TopOpt. The steady-state joint photon distribution $P(\mathbf{n}|\Phi)$ computed above is the marginal over time; a richer POVM uses the full *click-stream* waiting-time distribution $w(\tau_1, \tau_2, \ldots)$. Extracting the FI of this from the Keldysh framework requires two-time Green's functions and is an exciting direction — especially because pathwise MCWF naturally generates click-stream samples without computing the full joint distribution.

Open question

Non-Markovian inertial signal. The Sagnac coupling is instantaneous in the rotating frame, but mechanical resonances of the chip substrate add non-Markovian dynamics on ms timescales. Extending the Keldysh action to include mechanical modes (giving an optomechanical-Keldysh action) and computing the joint opto-mechanical FI for inertial parameters is a natural follow-on.

12 Summary and outlook

We have shown that the Keldysh QNM quantization framework of the companion manuscript can be *architecturally* extended to handle topology optimization of chip-scale quantum-comb inertial sensors:

1. **Above-threshold comb saddle:** reduces to the Lugiato–Lefever equation with explicit Sagnac coupling (Sec. 3.2).
2. **Comb-pair Bogoliubov:** each $\{+m, -m\}$ pair is a two-mode parametric squeezer with closed-form Lyapunov covariance, benchmarked against qutip to $< 1\%$ (Sec. 4).
3. **Free-space output:** input–output relation projects intracavity covariance to detector pixels; all n-point Gaussian observables computable from 2-point covariance (Sec. 5).
4. **Fisher information:** closed-form Gaussian formula Eq. (17) differentiable in design parameters (Sec. 6).
5. **Adjoint gradient:** reuses standard QNM adjoint plus local mode-product evaluations, total cost $\sim K + 1$ FDTD solves per gradient (Sec. 7).
6. **Worked example:** GaAs $R = 10\,\mu$m micro-ring, manoeuvre-rate sensitivity $\sim 10^{-2}$ rad/s/$\sqrt{\text{Hz}}$ at SQL, $\sim 10^{-3}$ with quantum enhancement (Sec. 8).

The framework is *ready for use* for Gaussian-comb inertial-sensor TopOpt at the 2-point level. Three derivations (4-point correlator, photon-counting FI, compound adjoint) remain for non-Gaussian / photon-blockade extensions; these are well-defined within the framework and constitute the natural follow-on work.

A Closed-form covariance at zero Sagnac

The Lyapunov equation $A_m C_m + C_m A_m^T + N_m = 0$ at $\Omega = 0$ admits the analytical solution

$$C_m = \frac{1}{4(1 - S_m^2)} \begin{pmatrix} 2 & 0 & 2S_m & 0 \\ 0 & 2 & 0 & -2S_m \\ 2S_m & 0 & 2 & 0 \\ 0 & -2S_m & 0 & 2 \end{pmatrix}, \qquad S_m = \frac{2\chi_m}{\kappa_m^{\text{eff}}}, \tag{38}$$

with eigenvalues $\{(1 \pm S_m)^{-1}/2\}$ each twice (two squeezed, two anti-squeezed eigenmodes).

B Sagnac shift for free-form geometry

For a generic 3D QNM $\tilde{\mathbf{f}}_m(\mathbf{r})$ in a structure rotating at angular velocity $\mathbf{\Omega}$:

$$\delta\tilde{\omega}_m = \frac{\omega_m}{c^2}\, \mathbf{\Omega} \cdot \text{Im} \int d^3r\, [\tilde{f}_m^* \times \nabla \tilde{f}_m] \Big/ \int d^3r\, \varepsilon |\tilde{f}_m|^2, \tag{39}$$

which reduces to Eq. (2) for an azimuthal mode of a ring resonator. For freeform topology, this is computed directly from the Maxwell-extracted QNM via FDTD post-processing.

C Worked Bogoliubov diagonalisation with Sagnac

The drift matrix A_m with both Sagnac splitting Ω_{eff} and parametric coupling χ_m has eigenvalues

$$\lambda_{\pm}^{(\sigma=\pm1)} = -\frac{\kappa_m^{\text{eff}}}{2} \pm \sqrt{\chi_m^2 - \Omega_{\text{eff},m}^2}, \quad \sigma = \pm, \tag{40}$$

each twice. The covariance is then explicitly reconstructed in the eigenbasis; for $\Omega_{\text{eff},m} < \chi_m$ (sub-Sagnac-locking), the two-mode squeezing structure persists; for $\Omega_{\text{eff},m} > \chi_m$, a phase transition to a Sagnac-dominated non-squeezed regime occurs. This bifurcation is itself a sensing resource (transition-edge detection for Ω).

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